



Name:

SCEGGS Darlinghurst

2004
Preliminary Course
Semester 2 Examination

Mathematics (Extension)

General Instructions

- Reading time – 5 minutes
- Working time – 1½ hours
- This paper has **four** questions
- Attempt **all** questions on the pad paper provided
- Write your name on every page
- Marks will be deducted for careless or badly arranged work
- Approved calculators may be used

Total marks – 60

- Attempt Question 1–4

Question	Communication	Reasoning	Total
1	/3	/3	/15
2	/4	/5	/15
3	/3	/6	/15
4		/11	/15
TOTAL	/10	/25	/60

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Instructions

- Attempt **all** questions on the pad paper provided
 - Write your name at the top of each page
 - Show all necessary working
 - Marks may be deducted for careless or badly arranged work
 - Approved calculators may be used
 - Begin each question on a new page
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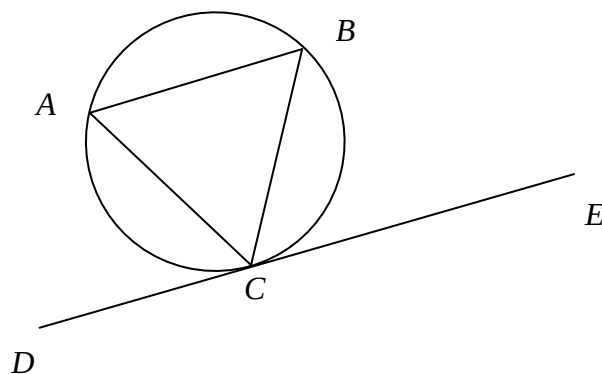
Question 1 (15 marks)**Marks**

- (a) Find the acute angle between the two lines $3x - 2y + 7 = 0$ and $7x + 5y - 6 = 0$. 3
Answer correct to the nearest minute.
- (b) Give α , β and γ are the roots of the equation $2x^3 - 3x^2 - 5x + 1 = 0$, find:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- (iii) $(\alpha + 1)(\beta + 1)(\gamma + 1)$ 2
- (c) Find the co-ordinates of the point P which divides the interval joining $A(2, -3)$ 2
and $B(4, 5)$ externally in the ratio 1 : 3.
- (d) Solve $\frac{3}{1-x} \leq 2$. 3

Question 1 continues on the next page

(e)

3



DE is a tangent to the circle at C .

$$\angle BCE = \angle ACD$$

Prove that AB is parallel to DE .

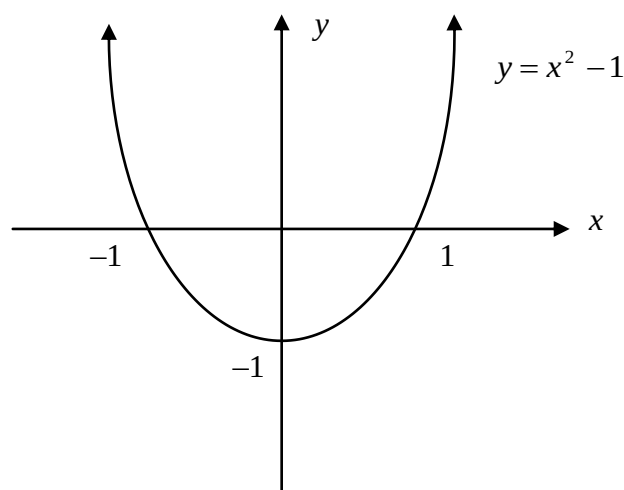
Question 2 (15 marks)

Marks

- (a) (i) Express $3\sin\theta - 2\cos\theta$ in the form $A\sin(\theta - \alpha)$ where $A > 0$ and α is an acute angle correct to the nearest minute. **3**

- (ii) Hence state the exact maximum value of $12 - 3\sin\theta + 2\cos\theta$. **1**

(b)



A student was given the curve $y = x^2 - 1$ as shown above. She was then asked to draw

$$y = \frac{1}{x^2 - 1}$$

- (i) Explain why she knew that asymptotes occur at $x = -1$ and $x = 1$. **1**
- (ii) Draw the curve $y = \frac{1}{x^2 - 1}$. **2**
- (c) (i) Prove that $x + 1$ is a factor of $P(x) = x^3 + 3x^2 - x - 3$. **1**
- (ii) Hence factorise $P(x)$ fully. **2**
- (iii) Solve $P(x) > 0$ **1**

Question 2 continues on the next page

Question 2 (continued)

Marks

- (d) Use the method of Mathematical Induction to prove that

4

$$9^{n+2} - 4^n$$

is divisible by 5 for all positive integers n .

Question 3 (15 marks)

Marks

- (a) Consider the polynomial

$$P(x) = x^3 - 3x^2 - 2x + 4$$

- (i) Show that it has a zero between $x = 3$ and $x = 4$. 2

- (ii) Taking an initial approximation of $x = 3.5$ and one application of Newton's Method, find a more accurate approximation to the zero correct to 3 significant figures. 2

- (b) Use the method of Mathematical Induction to prove that 5

$$2 + 5 + 8 + \dots + 3n - 1 = \frac{n(3n + 1)}{2}$$

for all positive integers n .

- (c) (i) Find the centre and radius of the circle whose equation is: 2

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

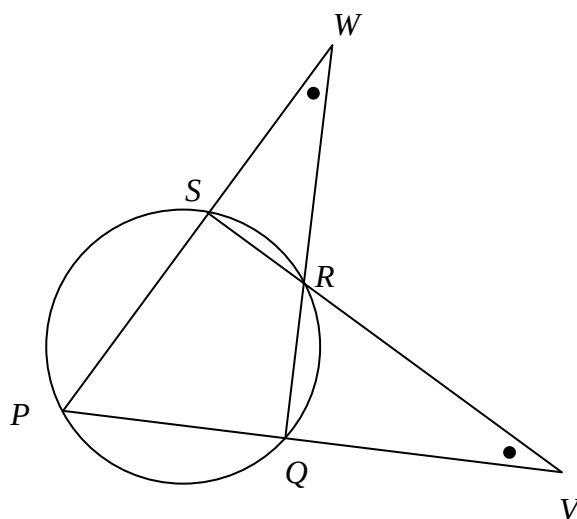
- (ii) Find, in terms of the constant k , the perpendicular distance from the centre to the line whose equation is $3x + 4y = k$. 2

- (iii) Hence find any values of k for which the line is a tangent to the circle. 2

Question 4 (15 marks)

Marks

(a)

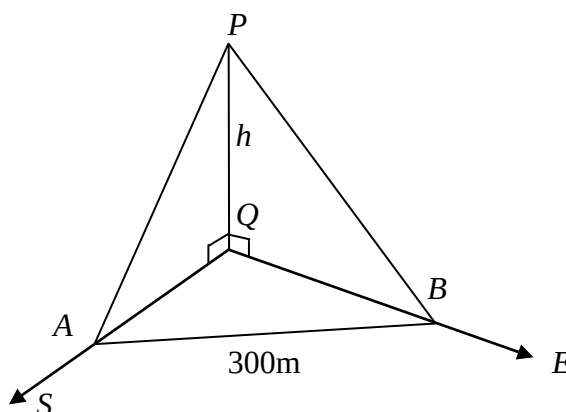


The points P , Q , R and S lie on the circumference of a circle. PS and QR are extended to W while PQ and SR are extended to V .

$$\angle PWQ = \angle SVP$$

- (i) Prove that $\angle PQR = \angle PSR$ 3
- (ii) Hence prove that PR is a diameter of the circle. 2

(b)

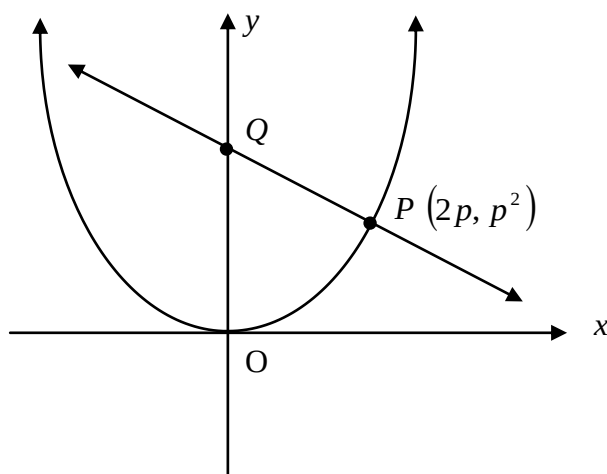


The points A and B are due South and East of a hill of height h m shown as PQ . A and B are 300 m apart. The angle of elevation of the top of the hill P from A is 30° and from B is 60° .

- (i) Prove that $AQ = \sqrt{3} h$. 1
- (ii) Hence prove that $h = 30\sqrt{30}$ metres. 3

Question 4 continues on the next page

(c)



$P(2p, p^2)$ is a point on the parabola $4y = x^2$.

- (i) Prove that the equation of the normal to the parabola at P has the equation 2

$$x + py = p^3 + 2p$$

- (ii) The normal meets the axis of the parabola at Q . Find the co-ordinates of Q . 1

- (iii) Find the co-ordinates of R , the midpoint of PQ . 1

- (iv) Show that the locus of R is a parabola and find its vertex. 2

End of paper

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① a) $2y = 3x + 7$ $5y = -7x + 6$

$$m_1 = \frac{3}{2}$$

$$m_2 = -\frac{7}{5}$$

$$\tan \theta = \left| \frac{\frac{3}{2} + \frac{7}{5}}{1 - \frac{3}{2} \times \frac{7}{5}} \right| = \left| \frac{\frac{29}{10}}{-\frac{11}{10}} \right| = \frac{29}{11}$$

$$\theta = 69^\circ 14' \text{ (nearest minute)}$$

b) (i) $\alpha + \beta + \gamma = \frac{3}{2}$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{5}{2}$ $\alpha\beta\gamma = -\frac{1}{2}$

(iii) $(\alpha+1)(\beta+1)(\gamma+1) = (\alpha+1)(\beta\gamma + \beta + \gamma + 1)$
 $= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \beta\gamma + \alpha + \beta + \gamma + 1$
 $= -\frac{1}{2} - \frac{5}{2} + \frac{3}{2} + 1$
 $= -\frac{1}{2}$

c) $(2, -3) (4, 5) \quad -1 : 3$

$$x = \frac{2 \times 3 + 4 \times -1}{2} = 1$$

$$y = \frac{-3 \times 3 + 5 \times -1}{2} = -7$$

P is $(1, -7)$

d) $\frac{3}{1-x} \times (1-x)^2 \leq 2(1-x)^2$

$$3(1-x) \leq 2(1-2x+x^2)$$

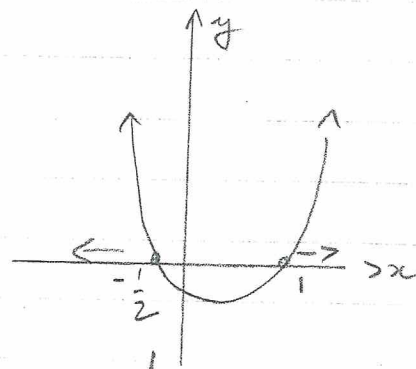
$$3-3x \leq 2-4x+2x^2$$

$$\therefore 2x^2 - x - 1 \geq 0$$

$$(2x+1)(x-1) \geq 0$$

$$\therefore x \leq -\frac{1}{2}, x \geq 1 \text{ but } x \neq 1$$

$$\therefore x \leq -\frac{1}{2}, x > 1 \text{ is solution}$$



(Reas)

e) $\angle BCE = \angle CAB$ (angle between tangent and chord equals angle in alternate segment)

But $\angle BCE = \angle ACD$ (given)

$$\therefore \angle ACD = \angle CAB$$

$\therefore AB \parallel DE$ (alternate angles are equal)

(comm)

② a) (i) $3 \sin \theta - 2 \cos \theta = A \sin \theta \cos \alpha - A \cos \theta \sin \alpha$

$$\therefore A \cos \alpha = 3 \quad \text{and} \quad A \sin \alpha = 2$$

$$\therefore A = \sqrt{13}$$

$$\cos \alpha = \frac{3}{\sqrt{13}}$$

$$\alpha = 33^\circ 24'$$

$$\therefore 3 \sin \theta - 2 \cos \theta = \sqrt{13} \sin(\theta - 33^\circ 24')$$

(ii) $-\sqrt{13} \leq \sqrt{13} \sin(\theta - 33^\circ 24') \leq \sqrt{13}$

(Rearr)

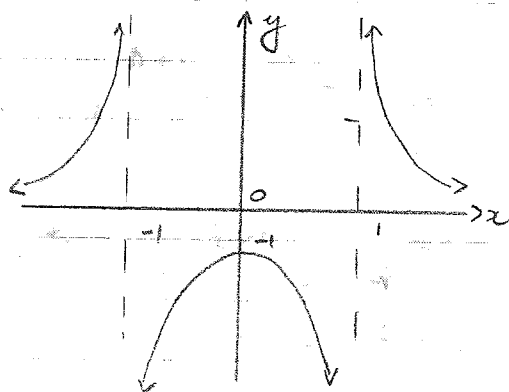
$\therefore$ maximum value is $12 + \sqrt{13}$.

b) (i) For the curve $y = x^2 - 1$, $xy = 0$ at $x = 1$ and $x = -1$.

$\therefore$ For the curve $y = \frac{1}{x^2 - 1}$, y is undefined at $x = 1$, $x = -1$

$\therefore$ asymptotes occur on $y = \frac{1}{x^2 - 1}$ at $x = 1$, $x = -1$. (comm.)

(ii)



2 (comm)

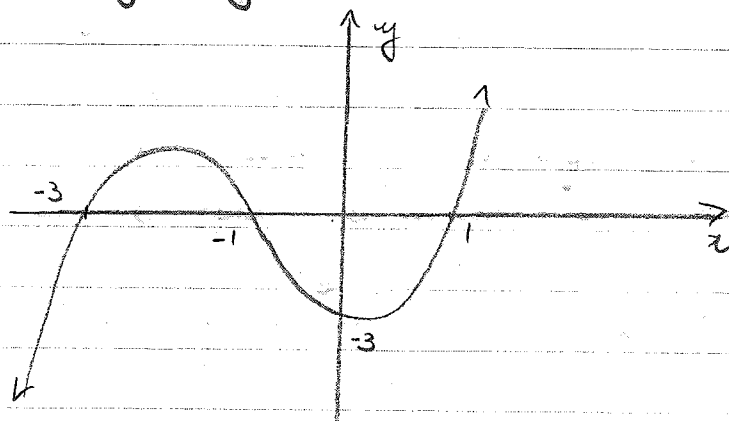
c) (i) $P(-1) = (-1)^3 + 3(-1)^2 + 1 - 3 = -1 + 3 + 1 - 3 = 0$

$\therefore x + 1$ is a factor of $P(x)$

$$\begin{array}{r}
 \text{(ii)} \quad \begin{array}{r}
 x^2 + 2x - 3 \\
 x+1 \overline{) x^3 + 3x^2 - x - 3} \\
 \underline{x^3 + x^2} \\
 2x^2 - x - 3 \\
 \underline{2x^2 + 2x} \\
 -3x - 3
 \end{array}
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x+1)(x^2 + 2x - 3) \\
 &= (x+1)(x+3)(x-1)
 \end{aligned}$$

(iii) Graphing $y = P(x)$



$P(x) > 0$ for $-3 < x < -1$ and $x > 1$ (com)

d) Assume true for $n = k$
i.e. that $9^{k+2} - 4^k$ is divisible by 5.

$\therefore$ Assume $9^{k+2} - 4^k = 5N$ where N is an integer.
Consider $n = k+1$

$$\begin{aligned}
 9^{k+3} - 4^{k+1} &= 9^{k+2} \times 9 - 4^{k+1} \\
 &= 9(5N + 4^k) - 4^{k+1} \\
 &= 45N + 9 \cdot 4^k - 4 \cdot 4^k \\
 &= 45N + 5 \cdot 4^k \\
 &= 5[9N + 4^k] \text{ which is}
 \end{aligned}$$

divisible by 5.

consider $n=1$, $9^3 - 4 = 729 - 4 = 725$ which is divisible by 5.

If the statement is true for $n=k$, it is also true for $n=k+1$. It is true for $n=1$ and thus is true for $n=2, 3, \dots$ etc. Thus true for all n positive integers.
4 (Reas)

(3) a) (i) $P(x) = x^3 - 3x^2 - 2x + 4$
 $P(3) = 3^3 - 3 \times 3^2 - 2 \times 3 + 4$
 $= -2 < 0$
 $P(4) = 4^3 - 3 \times 4^2 - 2 \times 4 + 4$
 $= 12 > 0$

since the sign has changed and $P(x)$ is continuous, there is a zero between $x=3$ and $x=4$. $\checkmark$ 2 (comm)

(ii) $P(3.5) = 3.5^3 - 3 \times 3.5^2 - 2 \times 3.5 + 4$
 $= 3.125$

$P'(3.5) = 3 \times 3.5^2 - 6 \times 3.5 - 2$
 $= 13.75$

$\therefore a_2 = 3.5 - \frac{3.125}{13.75}$
 $= 3.27 \text{ (3 s.f.)}$

b) Assume true for $n=k$
i.e. assume $2+5+8+\dots+3k-1 = \frac{k(3k+1)}{2}$

consider $n=k+1$

L.H.S. $= 2+5+8+\dots+3k-1+3k+2$
 $= \frac{k(3k+1)}{2} + 3k+2$
 $= \frac{3k^2+k+6k+4}{2}$
 $= \frac{3k^2+7k+4}{2}$

$$\begin{aligned}
 \text{L.H.S.} &= \frac{(k+1)(3k+4)}{2} \\
 &= \frac{(k+1)(3(k+1)+1)}{2} \\
 &= \text{R.H.S. if } n = k+1
 \end{aligned}$$

If $n=1$, L.H.S. = 2 4 (Reas)
 R.H.S. = 2

$\therefore$ If true for $n=k$, the statement is also true for $n=k+1$. It is true for $n=1$ and thus is true for $n=2, 3, \dots$
 $\therefore$ True for all n positive integers. 1 (con)

c) (i) $x^2 - 4x + 4 + y^2 + 6y + 9 = 12 + 4 + 9$
 $(x-2)^2 + (y+3)^2 = 25$
 centre is $(2, -3)$ radius is 5 units.

(ii) distance = $\left| \frac{3 \times 2 + 4 \times -3 - k}{5} \right|$
 $= \left| \frac{-6 - k}{5} \right|$ units.

(iii) if $\left| \frac{-6 - k}{5} \right| = 5$. ✓

$-6 - k = 25$

$k = -31$

or

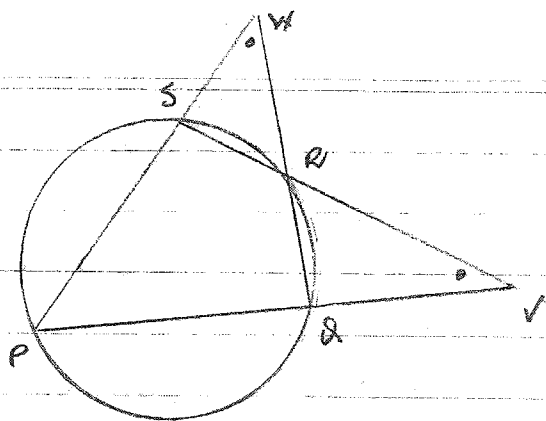
$6 + k = 25$

or

$k = 19$. ✓

$\frac{2}{1}$ (Reas)

④ a)



(i) $\angle QRV = \angle SRW$ (vertically opposite angles equal)

$\angle SWR = \angle RVQ$ (given)

$\therefore \triangle SWR \sim \triangle RVQ$ (equiangular)

$\therefore \angle RQV = \angle WSR$ (angles in similar Δ 's are equal)

$\angle PQR = 180^\circ - \angle RQV$ (straight angle is 180°)

$\angle PSR = 180^\circ - \angle WSR$ (" " " ")

$\therefore \angle PQR = \angle PSR$.

(ii) $\angle PQR + \angle PSR = 180^\circ$ (opp. angles in a cyclic quadrilateral are supplementary)

$\therefore 2\angle PQR = 180^\circ$

$\angle PQR = 90^\circ$

$\therefore PR$ is a diameter (if the angle is 90° it must be a semi circle)

(Reas) 5.

b) (i)

$\tan 30^\circ = \frac{h}{AQ}$

$\therefore AQ = \frac{h}{\frac{1}{\sqrt{3}}} = \sqrt{3}h$.

(Reas) 1.

(ii)

$\tan 60^\circ = \frac{h}{BQ}$

$BQ = \frac{h}{\sqrt{3}}$

using Pythagoras' Theorem,

$300^2 = 3h^2 + \frac{h^2}{3}$

$$90000 = \frac{10h^2}{3}$$

$$h^2 = 27000$$

$$h = \sqrt{27000} = \sqrt{900 \times 30}$$

$$= 30\sqrt{30} \text{ m}$$

3 Reas

c) (i) $y = \frac{x^2}{4}$

$$y' = \frac{2x}{4} = \frac{x}{2}$$

if $x = 2p$, $y' = \frac{2p}{2} = p$

equation of normal: $y - p^2 = -\frac{1}{p}(x - 2p)$

$$py - p^3 = -x + 2p$$

$$x + py = p^3 + 2p$$

(2)

(ii) if $x = 0$, $py = p^3 + 2p$

$$y = p^2 + 2$$

$\therefore$ it is $(0, p^2 + 2)$

(1)

(iii) midpoint: $(p, \frac{2p^2 + 2}{2})$ which is $(p, p^2 + 1)$ (1)

(iv) Let $x = p$, $y = p^2 + 1$

$$\therefore y = x^2 + 1$$

which is of the form of a parabola vertex $(0, 1)$

2 (Reas)